

Backup/Recovery

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Database Environment Overview

- Company size: 1500 employees, \$250M revenue, 10% annual growth.
- Database platforms:
 - Microsoft SQL Server (on-prem)
 - Third-party Cloud HR system
 - Systems: Accounting, Forecasting, Planning, Order Entry, Fulfillment, Data Warehouse
- Backup Strategy

SECTION 1 — Backup Schedule Table

Transactional Databases

Time	Type	Duration	Storage Location	Retention
12:00 AM	Full Backup	60 min	Local SAN + Cloud	30 days
Every 4 hrs	Differential	15 min	Local SAN	14 days
Every 15 min	Log	<1 min	Local disk	7 days

Data Warehouse

Day	Time	Type	Duration	Retention
Sunday	2 AM	Full	3 hrs	60 days
Daily	Midnight	Differential	25 min	30 days

Legacy System

Time	Type	Retention
Nightly	Full	90 days
Hourly	Log	14 days

Cloud Hr system

Backup Type	Frequency	Storage	Retention
Snapshots	Continuous	Vendor cloud	30 days
Exports	Daily	Internal secure storage	1 year

All SQL Server backups follow a 3-2-1 redundancy model. The primary backup copy is written to the local SAN in the primary data center. A secondary copy is automatically replicated to the disaster recovery site located in a geographically separate region. A third immutable copy is written to encrypted cloud object storage for ransomware protection. Transaction log backups are initially written to local disk and then moved to secure backup storage. Cloud HR backups are maintained by the vendor within multiple availability zones, and daily encrypted exports are

stored internally on secure file storage with restricted DBA access. Backup storage capacity is reviewed quarterly to accommodate 10% annual growth and increase database size.

Section 2- support protocols

Contact Structure and Escalation

Role	Responsibility	Contact Method	Availability
Lead DBA	Incident commander	Phone + Teams	24/7 on-call rotation
DBA #2	Backup operations	Teams + Email	Business hours + on call
DBA #3	Reporting/warehouse	Teams	Business hours
IT Director	Executive escalation	Phone	Emergency only
Cloud HR Vendor	SaaS support	Ticket portal + phone	SLA 24/7

System Priority Classification

Tier	Systems	Recovery Priority	RTO	RPO
Tier 1	Order Entry, Financial	Critical	1 hr	15 min
Tier 2	Planning, Forecasting	High	4 hrs	1 hr
Tier 3	Data Warehouse	Moderate	12 hrs	4 hrs
Tier 4	Legacy Archive	Low	24 hrs	24 hrs

Personnel Backup / Staffing Contingency

To prevent single-point-of-failure risk:

- All procedures are documented in a centralized knowledge base and operational runbooks
- Administrative credentials stored in secured enterprise password vault
- Quarterly cross-training between DBAs
- Access reviews performed monthly
- Backup job ownership shared among team members
- Credential escrow stored securely with IT Director

If DBA becomes unavailable (emergency, resignation, extended leave):

- Emergency credentials available to leadership
- Each critical system has at least two trained administrators
- Alternate DBA assumes responsibilities immediately
- Disaster recovery simulations conducted quarterly using backup personnel

Backup Storage Security

All backups are secured using enterprise security controls:

- AES-256 encryption at rest
- TLS encryption in transit
- Role-based access control
- MFA required
- Immutable storage copies
- Audit logging enabled
- Offsite geographic redundancy

SECTION 3 — Recovery Process

Local SQL Server Recovery Procedure

1. Identify failure scope (database, instance, storage, corruption).
2. Notify stakeholders and log incident.
3. Stop application services.
4. Verify backup integrity.
5. Set database to single user mode.
6. Restore most recent full backup.
7. Restore latest differential backup.
8. Restore transaction logs sequentially.
9. Run integrity checks:
10. Validate application functionality.

11. Return database to multi-user mode.
12. Resume services.
13. Document incident and resolution.

Disaster Recovery (Full Server Loss)

1. Provision new SQL Server host.
2. Install SQL Server + patches.
3. Restore system databases.
4. Restore production databases.
5. Reconfigure logins and permissions.
6. Redirect applications.
7. Verify connectivity.
8. Resume operations.

Cloud HR System Recovery

1. Open vendor support ticket.
2. Provide restore timestamp.
3. Vendor restores environment snapshot.
4. DBA validates data consistency.
5. Reconnect integrations.
6. Confirm functionality with HR team.

Validation Requirements

Recovery is considered successful only when:

- database online
- integrity checks pass
- application tests succeed
- users confirm access

- monitoring alerts clear

SECTION 4 — Metrics

Backup Performance Metrics

1. Backup success rate %
2. Restore success rate %
3. Average backup duration
4. Average restore duration
5. Failed job count
6. Backup size growth rate
7. Log backup delay time
8. Recovery SLA compliance %
9. Storage utilization %
10. Recovery test frequency
11. Time to detect failure
12. Vendor SLA compliance

Evaluation Techniques

- Monthly backup review
- Quarterly disaster recovery tests
- Automated alert monitoring
- Capacity forecasting analysis
- Trend analysis reports

Continuous Improvement Plan

If metrics fall below thresholds:

- investigate root cause
- adjust schedule

- increase frequency
- expand storage
- update documentation
- retrain staff

SECTION 5- Turnover Procedure

All production databases and system modifications follow a controlled implementation process based on risk classification. Low-risk changes such as index maintenance may be approved by a DBA and scheduled during standard maintenance windows, while higher-risk activities such as schema redesign, server migrations, or configuration changes require formal review, documented approval, and scheduled downtime notification. Prior to implementation, verified backups are performed to ensure rollback capability if a change fails. Requests are submitted through a standardized change form that documents scope, risk level, testing results, rollback plan, approvals, and scheduled deployment time. After implementation, validation testing is conducted and results are recorded. If issues occur, systems are restored from backup and the change is rescheduled. All turnover activities are logged for audit purposes, and documentation is updated whenever systems, procedures, or responsibilities change. Required forms include a change request form, approval record, deployment checklist, and post-implementation verification report, each completed at its respective stage of the process to ensure consistent, traceable, and controlled production updates.

Change Request form

FIELD	EXAMPLE ENTRY
CHANGE ID	001
DATE	02/20/2026
REQUESTED BY	Lead DBA
SYSTEM AFFECTED	Accounting Database
RISK LEVEL	High
DESCRIPTION OF CHANGE	Schema redesign to support new quarterly reporting fields
PRE-CHANGE BACKUP VERIFIED	Yes
ROLLBACK PLAN	Restore last full backup, apply latest differential, validate integrity

SCHEDULED DEPLOYMENT TIME	02/21/2026 1:00 AM
APPROVALS	IT Director, DBA Lead

Deployment Checklist

TASK	COMPLETED (Y/N)	NOTES
BACKUP VERIFIED	Y	Full + differential + log backups complete
USERS NOTIFIED	Y	Finance team notified 24h prior
DOWNTIME SCHEDULED	Y	Maintenance window 1:00–3:00 AM
CHANGE SCRIPTS VALIDATED	Y	Tested on staging
ROLLBACK SCRIPTS READY	Y	Stored in secure vault
POST-CHANGE MONITORING	Y	Alerts configured for 24h

Post-Implementation Verification Report

FIELD	EXAMPLE ENTRY
CHANGE ID	001
VERIFIED BY	DBA #2
VERIFICATION DATE	02/21/2026
FUNCTIONALITY TESTS	Passed: Reports generate correctly, queries return expected results
BACKUP INTEGRITY POST-CHANGE	Verified

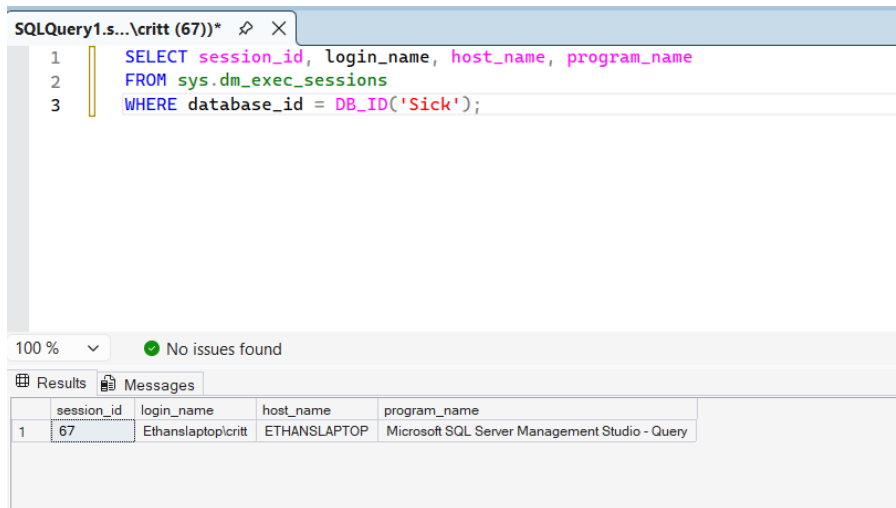
ISSUES FOUND	None
RESOLUTION	N/A
NOTES	System returned to production and monitoring normal

Backout Plan

FIELD	EXAMPLE ENTRY
SYSTEM	Accounting Database
BACKUP LOCATION	D:\Backups\Accounting.bak
RESTORE METHOD	Full restore + latest differential + transaction logs
ESTIMATED RESTORE TIME	45 min
NOTIFICATION PLAN	DBA team + Finance team
TRIGGER CONDITIONS	Errors in reports, failed schema changes, corruption detected
STEPS	1. Set database to single-user mode 2. Restore full backup 3. Apply differential/log backups 4. Validate integrity 5. Set multi-user mode

SECTION 6 — Identify Logged-In Users & Kill Session

See Active Users Connected- shows the connected users to sick database.



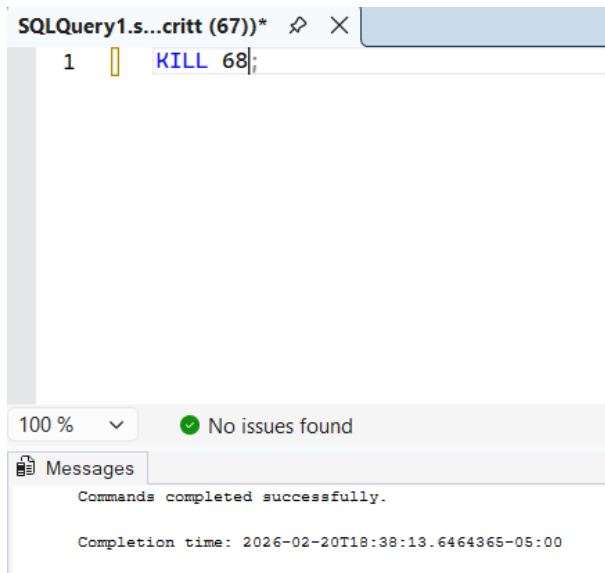
The screenshot shows a SQL query window titled "SQLQuery1.s...critt (67))*" with the following SQL code:

```
1 SELECT session_id, login_name, host_name, program_name
2 FROM sys.dm_exec_sessions
3 WHERE database_id = DB_ID('Sick');
```

Below the query window, the "Results" tab is active, displaying a table with the following data:

	session_id	login_name	host_name	program_name
1	67	Ethanslaptop\critt	ETHANSLAPTOP	Microsoft SQL Server Management Studio - Query

Kill User Session- in this case it kills the session of user id 68.



The screenshot shows a SQL query window titled "SQLQuery1.s...critt (67))*" with the following SQL code:

```
1 KILL 68;
```

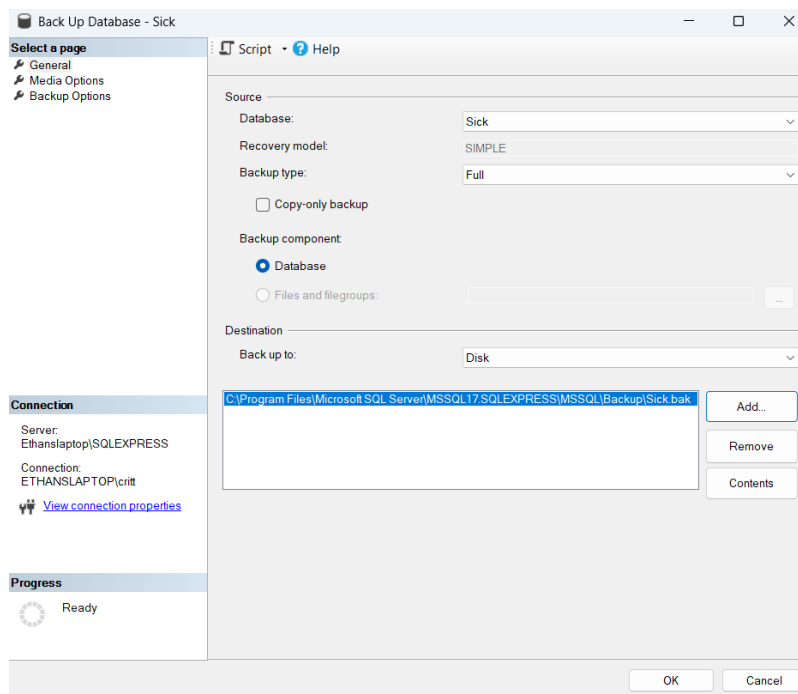
Below the query window, the "Messages" tab is active, displaying the following messages:

```
Commands completed successfully.
Completion time: 2026-02-20T18:38:13.6464365-05:00
```

SECTION 7- BU/Recover w/ Wizard

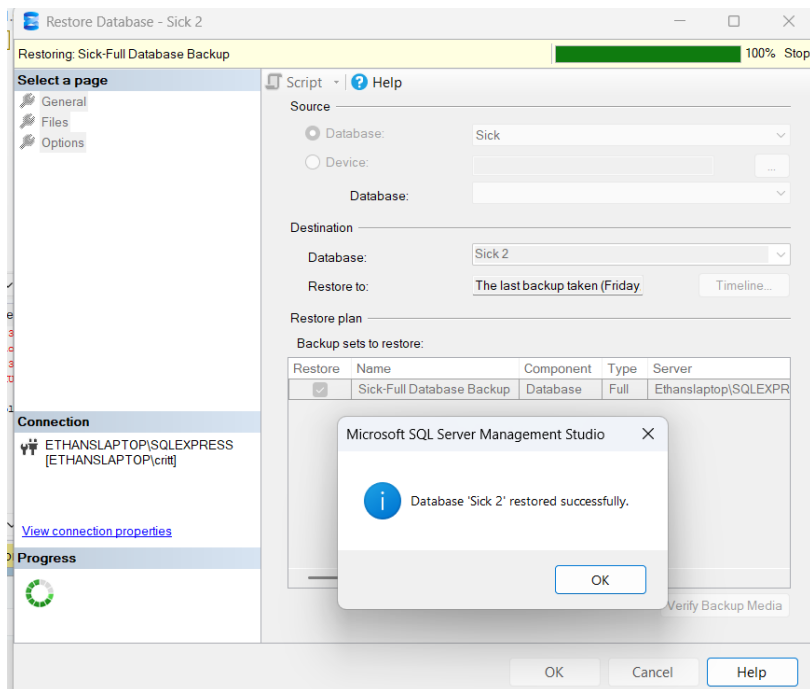
Backup a Database

- Right-click database
- Select Tasks
- Click Backup
- Choose Full
- Select destination
- Click OK



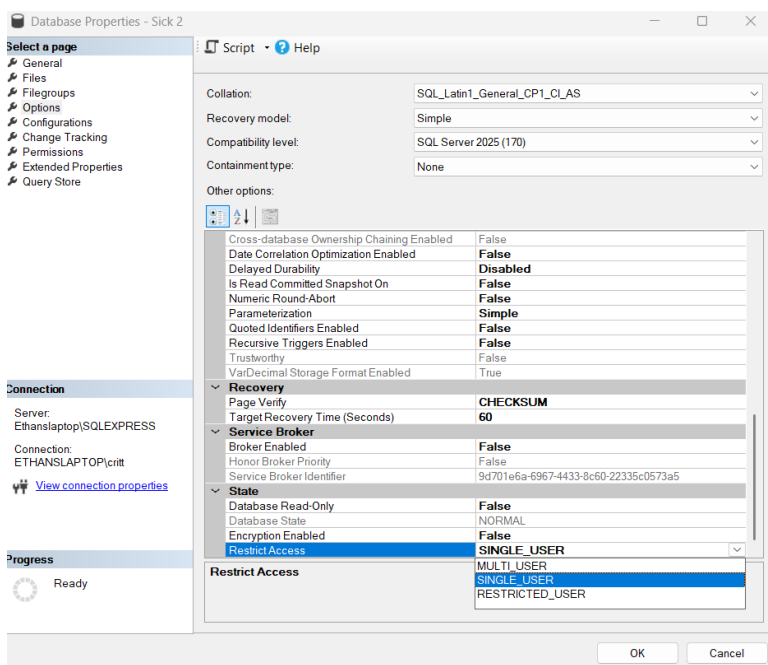
Restore a Database

- Right-click Databases
- Select Tools - Restore Database
- Choose backup file
- Enter new database name
- Click OK



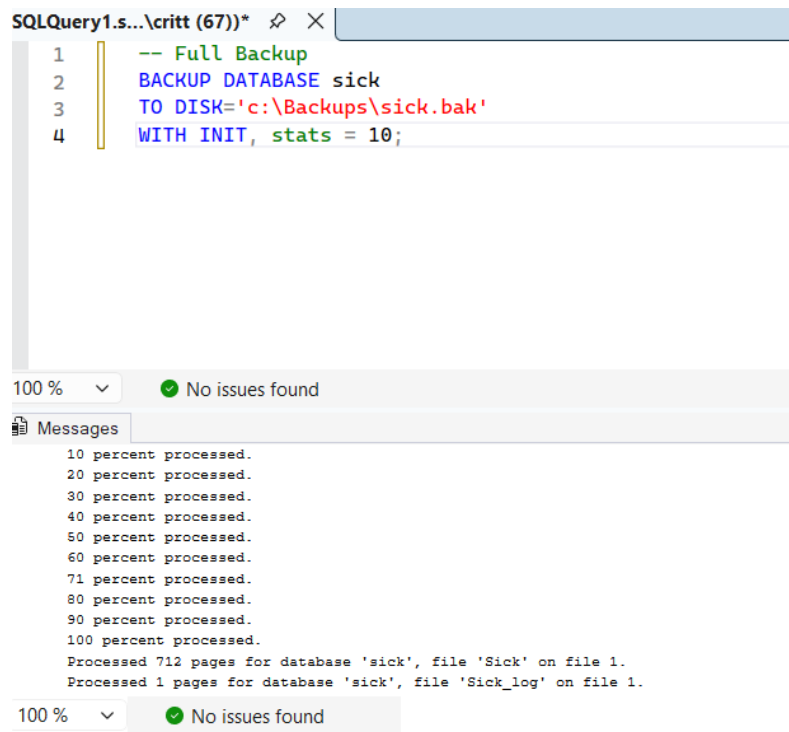
Single to Multi Access Database

- Right Click Database
- Properties – options
- Scroll down click Restrict Access choose single user or multiuser.
- Hit ok



SECTION 8- BU/Recover w/ scripts

Backup with scripts



The screenshot displays the SQL Server Enterprise Manager interface. At the top, a query window titled 'SQLQuery1.s... \critt (67))' contains a T-SQL script for a full backup of the 'sick' database. The script is as follows:

```
1  -- Full Backup
2  BACKUP DATABASE sick
3  TO DISK='c:\Backups\sick.bak'
4  WITH INIT, stats = 10;
```

Below the query window, the 'Messages' pane shows the execution progress of the backup. It indicates that the backup is 100 percent processed and provides details on the pages processed for the database and its log files.

100 % No issues found

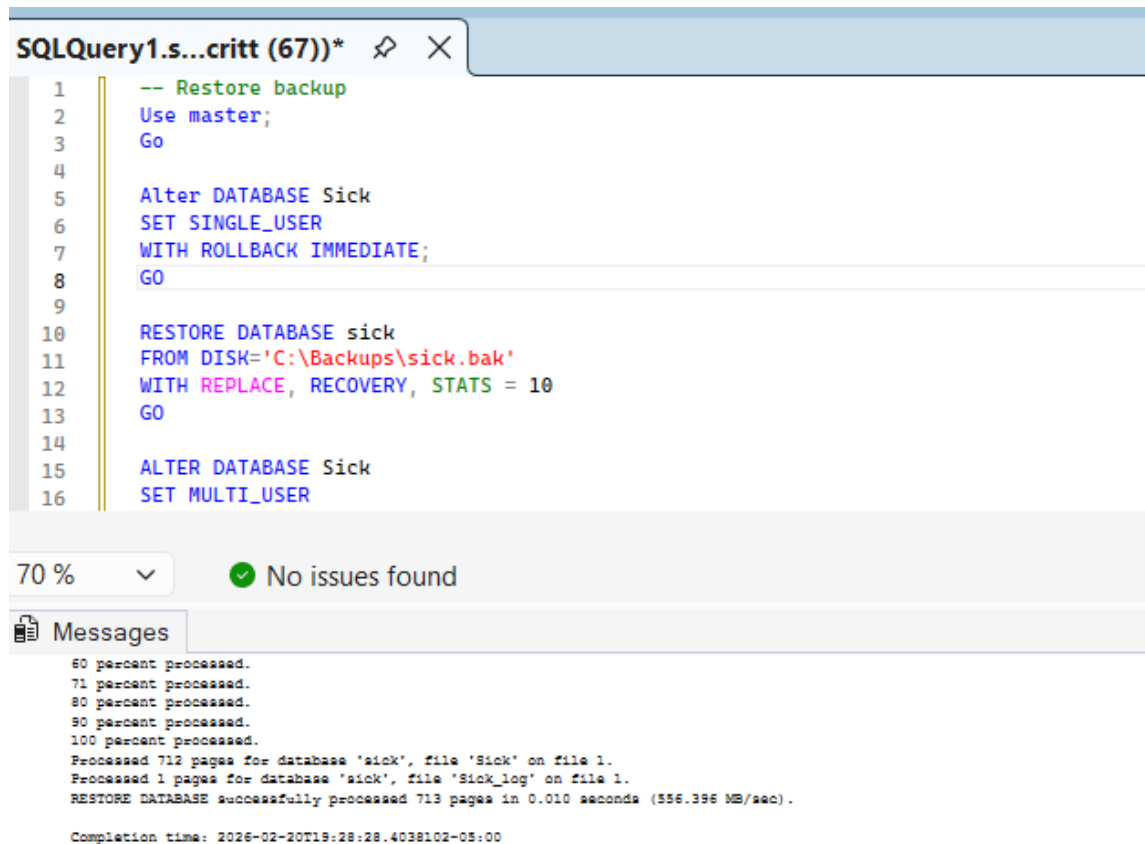
Messages

```
10 percent processed.
20 percent processed.
30 percent processed.
40 percent processed.
50 percent processed.
60 percent processed.
71 percent processed.
80 percent processed.
90 percent processed.
100 percent processed.
Processed 712 pages for database 'sick', file 'Sick' on file 1.
Processed 1 pages for database 'sick', file 'Sick_log' on file 1.
```

100 % No issues found

This script performs a full backup of the “sick” database to the specified disk location. The INIT option overwrites any existing backup file at that location. The STATS = 10 option displays progress updates in 10% increments during the backup process.

Restore with scripts



The screenshot shows the SQL Server Enterprise Manager interface. At the top, a tab titled 'SQLQuery1.s...critt (67))' is open. Below the tab, a script window displays the following T-SQL code:

```
1  -- Restore backup
2  Use master;
3  Go
4
5  Alter DATABASE Sick
6  SET SINGLE_USER
7  WITH ROLLBACK IMMEDIATE;
8  GO
9
10 RESTORE DATABASE sick
11 FROM DISK='C:\Backups\sick.bak'
12 WITH REPLACE, RECOVERY, STATS = 10
13 GO
14
15 ALTER DATABASE Sick
16 SET MULTI_USER
```

Below the script window, a status bar shows '70 %' and a green checkmark icon with the text 'No issues found'. At the bottom, a 'Messages' window is open, displaying the following output:

```
60 percent processed.
71 percent processed.
80 percent processed.
90 percent processed.
100 percent processed.
Processed 712 pages for database 'sick', file 'Sick' on file 1.
Processed 1 pages for database 'sick', file 'Sick_log' on file 1.
RESTORE DATABASE successfully processed 713 pages in 0.010 seconds (556.396 MB/sec).

Completion time: 2026-02-20T19:28:28.4038102-05:00
```

- **USE master;**
Switches the context to the master database so the target database can be modified.
- **SET SINGLE_USER WITH ROLLBACK IMMEDIATE;**
Forces the database into single-user mode and immediately terminates any active connections.
- **RESTORE DATABASE**
Restores the Sick database from the specified backup file.
- **WITH REPLACE**
Allows SQL Server to overwrite the existing database.
- **WITH RECOVERY**
Brings the database online and makes it operational after the restore.
- **STATS = 10**
Displays progress messages in 10% increments during the restore process.
- **SET MULTI_USER**
Returns the database to normal multi-user access after the restore completes.